

Enterprise Operational Performance & Resource Predictor

1. Introduction

The **Enterprise Operational Performance & Resource Predictor** is a Power BI-based healthcare analytics project designed to monitor operational resources, analyze performance, identify resource deficits, and provide predictive insights.

The dashboard combines operational KPIs, interactive visualizations, geographical mapping, decomposition analysis, time-intelligence DAX measures, predictive analysis, and automated resource alerts.

2. Project Objective

The main objective of this project is to build a multi-tier dashboard that helps identify operational resource shortages and performance issues.

The project aims to:

- Monitor available and required resources.
- Identify resource deficits.
- Analyze resource utilization and performance.
- Compare regions, departments, resource types, and facilities.
- Visualize facility locations using geographical coordinates.
- Analyze historical resource trends.
- Estimate future resource demand.
- Generate automated resource alerts.
- Provide interactive filtering for operational analysis.

3. Dataset Description

A simulated healthcare operational dataset was used for this project.

The dataset contains operational information including:

- Date
- Facility
- Region
- Department
- Resource Type

- Available Resources
- Required Resources
- Patient Demand
- Utilization %
- Performance %
- Latitude
- Longitude
- Resource Deficit
- Alert Status

The dataset was imported into **Microsoft Power BI** for analysis and visualization.

4. Tools & Technologies

The following tools and technologies were used:

- **Microsoft Power BI Desktop**
- **DAX (Data Analysis Expressions)**
- **Microsoft Excel**
- Data Visualization
- Time-Series Analysis
- Spatial Mapping
- Predictive Analysis

5. Data Preparation

The dataset was loaded into Power BI and the data types were checked.

The Date column was configured as a **Date** data type.

The dataset was then used to create measures for:

- Available resources
- Required resources
- Resource deficit
- Utilization

- Performance
- Historical resource trends
- Predictive demand
- Automated alerts

6. DAX Measures

Several DAX measures were created to support operational and predictive analysis.

Total Available Resources

Total Available Resources =

SUM (Sheet1[Available Resources])

Total Required Resources

Total Required Resources =

SUM (Sheet1[Required Resources])

Total Resource Deficit

Total Resource Deficit =

SUM (Sheet1[Resource Deficit])

Average Utilization

Average Utilization =

AVERAGE (Sheet1[Utilization %])

Average Performance

Average Performance =

AVERAGE (Sheet1[Performance %])

Previous Month Deficit

Previous Month Deficit =

CALCULATE (

[Total Resource Deficit],

DATEADD(Sheet1[Date], -1, MONTH)

)

Deficit Growth %

Deficit Growth % =

DIVIDE (

$$\frac{[\text{Total Resource Deficit}] - [\text{Previous Month Deficit}]}{[\text{Previous Month Deficit}]}$$

0

) * 100

30-Day Moving Average

30 Day Moving Average =

AVERAGEX (

DATESINPERIOD (

Sheet1[Date],

MAX(Sheet1[Date]),

-30,

DAY

),

[Total Resource Deficit]

)

Predicted Resource Demand

Predicted Resource Demand =

AVERAGEX (

DATESINPERIOD (

Sheet1[Date],

MAX(Sheet1[Date]),

-30,

DAY

),

CALCULATE (SUM (Sheet1[Patient Demand]))

)

Automated Resource Alert

Resource Alert =

SWITCH (

TRUE (),

[Total Resource Deficit] > 15, "Critical",

[Average Utilization] > 90, "Critical",

[Average Utilization] >= 75, "Warning",

"Healthy"

)

7. Dashboard Design

The dashboard was divided into **four pages** to keep the analysis organized and easy to understand.

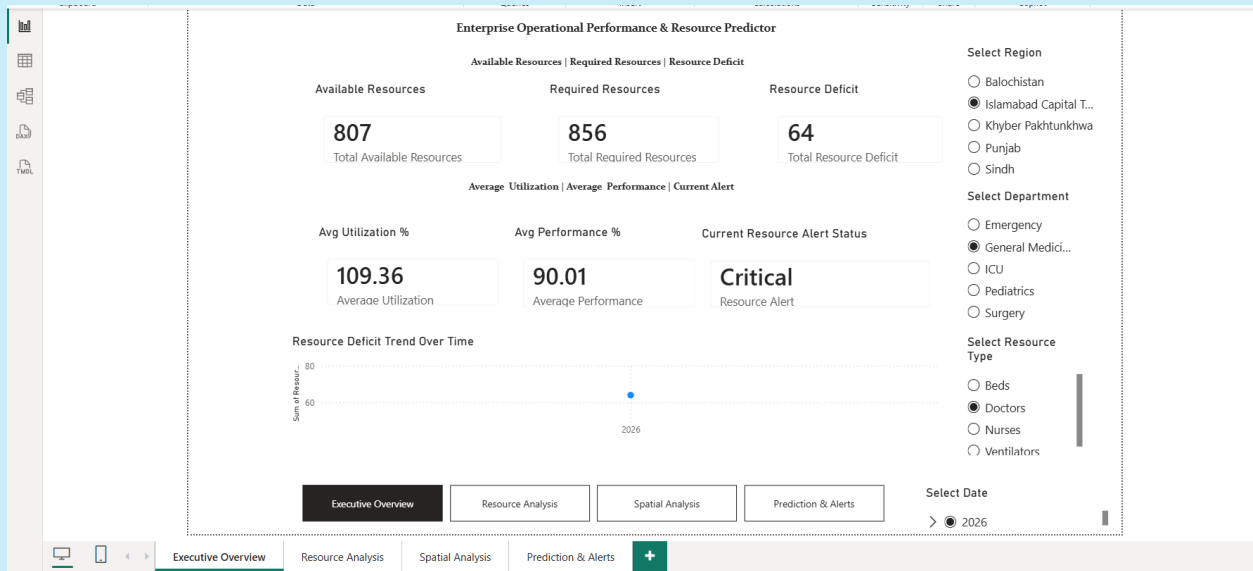
Page 1 — Executive Overview

The Executive Overview provides a summary of the overall operational situation.

It contains:

- Total Available Resources
- Total Required Resources
- Total Resource Deficit
- Average Utilization
- Average Performance
- Current Resource Alert Status
- Resource Deficit Trend Over Time

Screenshot:



Page 2 — Resource Analysis

The Resource Analysis page focuses on identifying resource deficits across operational dimensions.

Decomposition Tree

The Decomposition Tree analyzes:

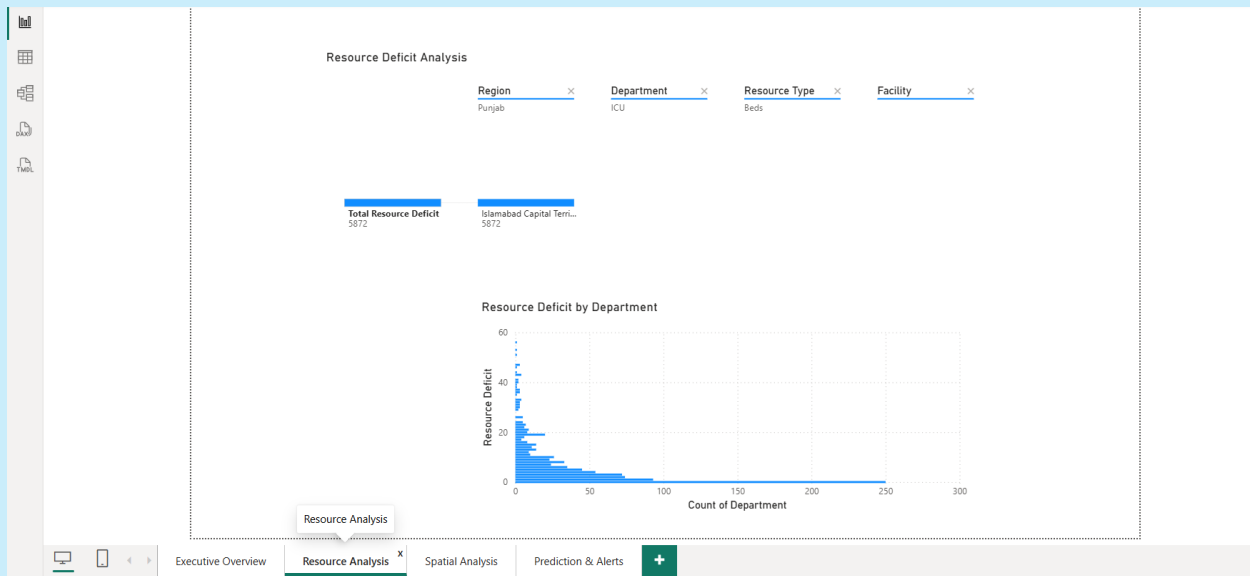
Region → Department → Resource Type → Facility

It helps identify which operational areas contribute most to the overall resource deficit.

Additional visuals include:

- Resource Deficit by Department
- Resource Utilization by Department

Screenshot:



Page 3 — Spatial Resource Analysis

The Spatial Resource Analysis page uses geographical coordinates to visualize healthcare facilities.

The map uses:

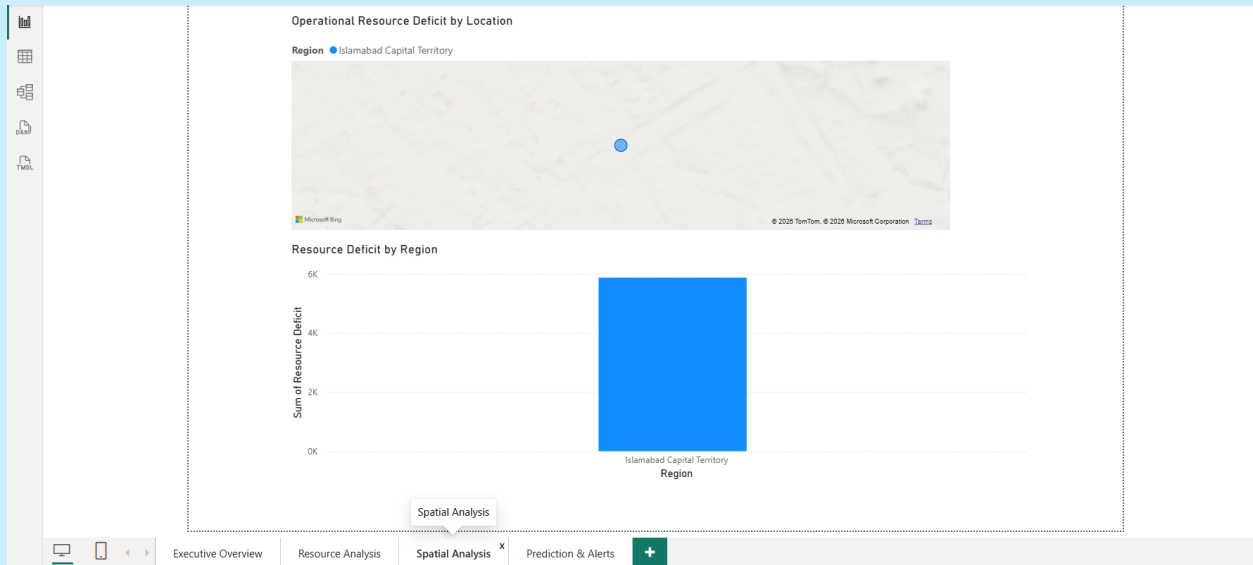
- Latitude
- Longitude
- Facility
- Region
- Resource Deficit

Additional visualizations include:

- Resource Deficit by Region
- Performance by Facility

This provides geographical context for operational resource shortages.

Screenshot:



Page 4 — Prediction & Automated Alerts

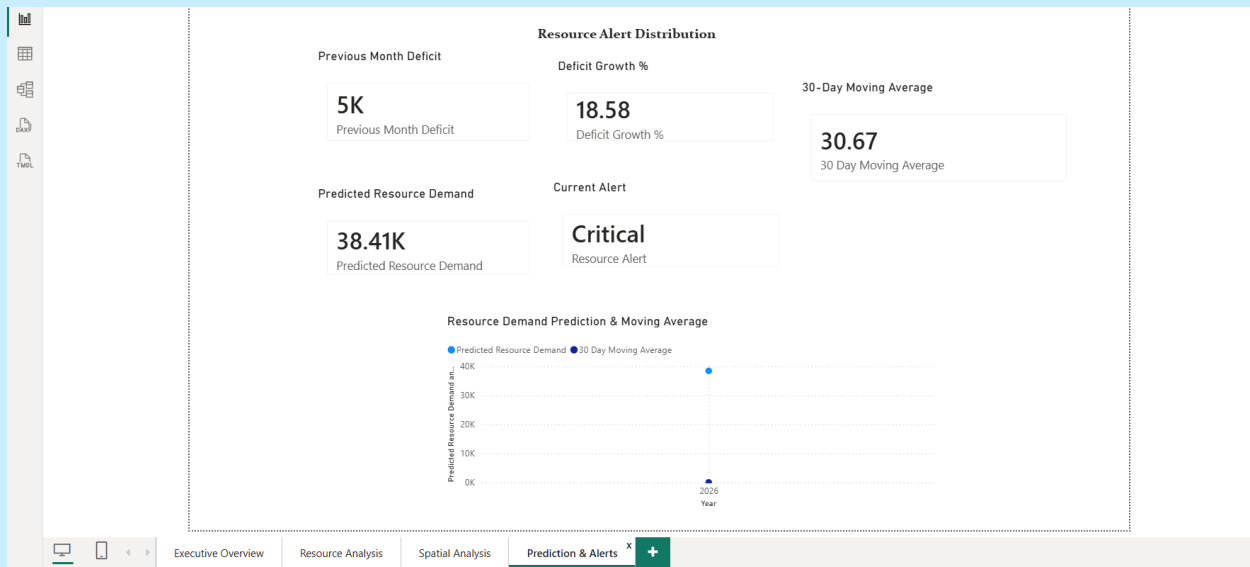
The Prediction & Automated Alerts page focuses on time-based and predictive analysis.

It contains:

- Previous Month Deficit
- Deficit Growth %
- 30-Day Moving Average
- Predicted Resource Demand
- Current Resource Alert
- Resource Demand Prediction & Moving Average
- Resource Alert Distribution

The moving average provides a historical trend, while the predicted demand measure provides an estimate based on recent demand patterns.

Screenshot:



8. Interactive Features

The dashboard includes interactive slicers for:

- Region
- Department
- Resource Type
- Date

The slicers were synchronized across the dashboard pages so that filtering information on the main dashboard can affect the analysis across different pages.

Page navigation was also added to allow users to move between:

Executive Overview → Resource Analysis → Spatial Analysis → Prediction & Alerts

9. Automated Alerts

An automated alert measure was created to identify operational conditions.

The alert logic classifies resource conditions into:

- **Critical**
- **Warning**
- **Healthy**

Critical conditions are triggered when resource deficits exceed the defined threshold or utilization becomes very high.

Warning conditions indicate elevated utilization levels that may require monitoring.

Healthy conditions indicate that utilization remains below the warning threshold.

10. Key Insights

The dashboard enables users to:

- Identify areas with resource shortages.
- Compare resource requirements with availability.
- Determine departments with higher resource deficits.
- Analyze facility-level performance.
- Identify geographical resource distribution.
- Monitor changes in resource deficits over time.
- Compare recent demand with historical demand patterns.
- Detect potentially critical operational conditions.

11. Conclusion

The **Enterprise Operational Performance & Resource Predictor** demonstrates how Power BI can be used to transform healthcare operational data into an interactive analytical dashboard.

The project integrates KPI monitoring, Decomposition Tree analysis, spatial mapping, time-intelligence DAX calculations, predictive resource analysis, interactive slicers, and automated alerts.

The resulting dashboard provides a centralized view of operational performance and helps users identify potential resource shortages and areas requiring attention.

12. Project Files

The project contains:

- Power BI dashboard file (pbix)
- Excel dataset (xlsx)
- Project documentation (README.md)

13. Project Information

Project Title: Enterprise Operational Performance & Resource Predictor

Domain: Healthcare Analytics

Technology: Microsoft Power BI

Project Type: Mini Project

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